

A FRAMEWORK FOR THE DESIGN AND DEPLOYMENT OF LARGE-SCALE LPWAN NETWORKS FOR SMART CITIES APPLICATIONS

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ABSTRACT

The Internet of Things (IoT) enables the connectivity of massive numbers of sensors to facilitate the delivery of immense sets of data. This valuable data promotes higher productivity, effectiveness, and sustainability. With the accelerating global urbanization, many cities and suburbs are expected to face increasing challenges related to urban planning, resources and livability. These challenges stimulate new opportunities for communities and governments to collaboratively incubate new planning and development strategies relying on the insightful IoT data. This concept is commonly referred to as smart cities. Accordingly, a wave of smart cities implementations is currently emerging. However, the underlying IoT network deployments are developed in silos, and thus require more harmonization. In this paper, we present a design and deployment framework for large-scale low power wireless access networks (LPWAN) in order to harmonize and guide the end-to-end process. This framework is derived from the acquired expertise and learned lessons from the largest open-access LPWAN deployment in Australia. The framework includes various aspects related to the state-of-the-art design techniques, smart cities use-case scenarios, project development and implementation, network management, and finally a robust performance testing mechanism for quality of service assurance. All these components were put into rigorous verification and testing under this realistic large-scale IoT network deployment.

INTRODUCTION

More than 68 percent of the world's population is projected to live in urban cities by 2050 [1] almost doubling the number of their residents. Growth of this magnitude would certainly provide many economic opportunities, but will equally present several challenges including higher traffic congestion, more pressure on natural resources, and increased levels of CO₂ emissions and other pollutants. This projection has led many governments around the world to adopt a transformative smart cities vision to address these arising challenges. The renewed focus on smarter cities will not only lead to a better utilization of infrastructure and improvements in energy efficiency and services, but will also ensure that cities remain globally competitive and attractive. Smart cities include many application scenarios which have particularly been of a great interest to local governments [2] ranging from smart parking, pollution monitoring, occupancy estimation, smart lighting, smart metering, asset tracking, waste management to water-level monitoring. Among the many smart cities technology drivers is the Internet of Things (IoT) that plays a major role in shaping smart cities and their future road-map [3]. At the heart of the smart city model, a massive number of IoT sensors collect data from their surroundings to enable back-end analytics [4]. These analytics draw insightful understanding of the current state of a city and project different solutions based on the evolving Artificial Intelligence (AI).

To cater to battery-powered IoT wireless devices, several technologies have been proposed, collectively referred to as low power wide area networks (LPWAN) aiming for an extend-

ed battery life of several years. A widely deployed LPWAN technology is LoRaWAN, supported by the LoRa Alliance which is an association of more than 500 member companies. LoRaWAN utilizes a proprietary modulation and coding scheme based on a chirp spread spectrum (CSS) technique which spreads over a larger radio spectrum for better interference immunity. Signals with different spreading factors can coexist in the channel. Moreover, increasing the spreading factor enhances the communication link budget, making the signal more resilient to remote deployments. However, a higher spreading factor also lowers the data rate, leading to longer transmission time and increasing energy drainage [5].

Over the past few decades, the deployment of large-scale networks was almost always associated with large telco operators. However, with the advent of simpler LPWAN technologies, emergent network deployments can be carried out by smaller sized new comers to the market or by the end-users themselves. Accordingly, the harmonization and integration of such deployments becomes of utmost importance to allow the LPWAN to provide a seamless continuum and avoid unnecessary capital investment. The harmonization needs to be accounted for not only in the technology, but also in the way the network is designed, tested, and in the quality assurance. Ongoing standardization efforts for IoT data network infrastructures exist [6]; nevertheless, standardization for the network design and implementation phases is still underexplored.

In this article, we present an end-to-end framework based on the lessons learned from the deployment of a large-scale LPWAN for smart cities application in Melbourne, Australia. The framework provides a blueprint to guide the process of network development all the way from the use-case scenario creation, to the radio design methodology, network architecture, network management, implementation and finally the testing and quality assurance. The framework is developed and put into action through the development of the Northern Melbourne Smart Cities Network (NMSCN) project, which is the largest single deployment of an open-access LPWAN network in Victoria, Australia. The NMSCN network has already integrated

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Digital Object Identifier: 10.1109/IOTM.001.2000179

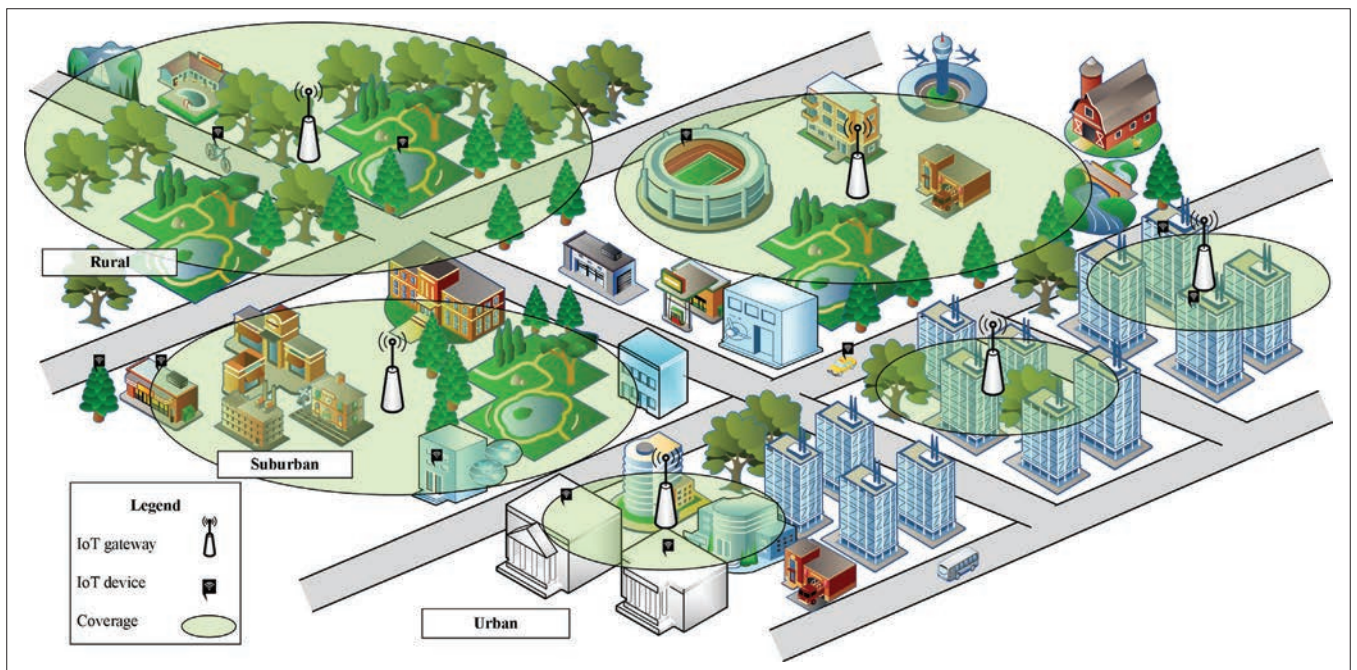


FIGURE 1. IoT Smart city concept in different urban environments where radio coverage largely depends on the interference levels and the propagation conditions.

48 LoRaWAN gateways and 294 sensors across the five contributing local council areas. The framework underscores the network design, radio frequency testing, connectivity testing, and quality assurance of the network. Ultimately, the framework is aimed to help network planning engineers, researchers and decision makers alike to gain insightful understanding of the challenges and opportunities of large scale deployments, as well as facilitate better harmonization in future IoT networks and expansions.

NORTHERN MELBOURNE SMART CITIES NETWORK

The Australian Government launched the smart cities plan in 2016 as a long-term framework to support the growth of the cities and suburbs. The aim of this framework is to achieve higher sustainability, better productivity, and more liveable cities. One of the main pillars of this plan is the use of Smart Technology to drive investment and policies [7]. Smart Technology is a broad term, but the plan has further emphasized the importance of open and real time data for informing better policies and planning. To kick start the plan, AUD \$50 million (around USD \$35 million) were awarded in the form of competitive grants to the bidding councils.

The bid to build the NMSCN network was successfully selected and the project was awarded AUD \$1.5 million to build an open IoT network covering spots across the five contributing councils [8]. The network is purposed to gather sensor data to support further analytics required to monitor and improve the efficiency of existing services and to ultimately support the potential of new service delivery. The NMSCN is overarching a diverse combination of inner city, interface, and rural councils as illustrated in Fig. 1. To offer services to their respective communities, the contributing local councils maintain many assets such as public buildings, parks, footpaths, roads, playgrounds, public toilets, and public BBQs, among many others. However, prior to the Australian Government initiative, councils invested a considerable amount of time and money in the construction and maintenance of public assets with little data showcasing their actual usage. Hence, one of the main objectives of the NMSCN project is to enable the collection of data on the usage of public assets by the community. This enables councils to promote the following objectives:

- Offer new services such as advising the community about air quality and water levels in the local creeks.
- Develop trigger-based alert systems for both public safety and internal staff use.
- Understand community demands and better allocate assets and services.
- Invest in future services based on data trends and community behavior prediction.
- Motivate innovation by allowing open-access to the network by the public.

The technical objective of the NMSCN project is to establish an open-access LoRaWAN network consisting of an initial roll out of 48 gateways and 294 sensors. Nevertheless, the ultimate goal of the project is for the network to be scalable and engulf more sensors managed by the councils or the community in the short-term future. The choice of the sensors was based on the overwhelming interest shown by the councils in five use-cases during the initial consultation phases of the project:

Waste management sensors to capture the bin usage and determine trends of usage over time across the community.

Water level sensors to monitor the levels of water in creeks and alert councils in the case of flooding and to monitor levels in regional area fire fighting water towers.

Occupancy estimation sensors to anonymously capture data of pedestrian traffic in shopping strips, public parks and other public areas of interest.

Environmental sensors as a new service to understand localized air quality, humidity, temperature and other environmental parameters.

Asset tracking sensors to better track mobile council services such as street sweepers, mowers and council vehicles to better allocate these resources.

In total, eight organizations were working together to deliver this largest open-access IoT network in Victoria. The project is a manifestation of the collaboration work between councils, universities and industry. Please refer to the official web page of the project for more details about the contributing partners. Five Councils came together to be part of the NMSCN project: City of Whittlesea Council (lead council), Banyule City Council, Moreland City Council, Mitchell Shire Council and Nillumbik Shire Council. Royal Melbourne Insti-

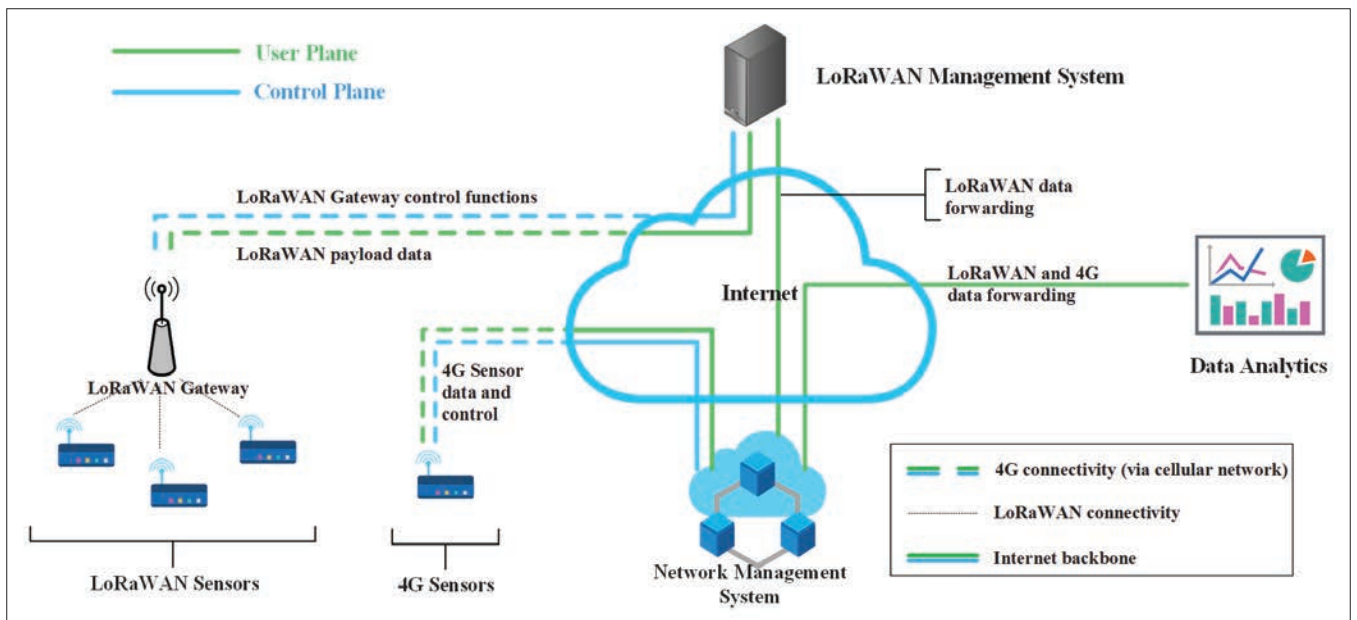


FIGURE 4. Cellular IoT model network architecture adopted in NMSCN.

tute of Technology (RMIT) University is a partner responsible for grant preparation, network design, testing and quality assurance. La Trobe University is another partner responsible for the data analytics and visualization. Minnovation Australia is the chosen partner responsible for the supply of the sensors and gateways along with all associated installation services as well as hardware maintenance and support. This project attracted two prestigious industry awards from Municipal Association of Victoria (MAV) Technology Awards for Excellence 2020, and IoT Alliance Australia (IoTAA) Smart Cities Award for 2020.

DESIGN FRAMEWORK

NETWORK ARCHITECTURE

In principle, the adopted layout in NMSCN follows the typical cellular network topology, where IoT devices communicate directly with their serving LoRaWAN gateways. The network primarily utilizes LTE backhaul over commercial telecom operators. In some cases, where the LoRaWAN coverage becomes inadequate, specifically when the IoT devices are quite sparse, 4G LTE radio modules are utilized such that the IoT traffic is handled directly by the commercial telecom operators. Figure 2 illustrates the end-to-end connection and highlights the adopted network architecture in NMSCN which is constituted of the following:

IoT Devices: having different types of sensors as indicated in the previous section, equipped with the radio network access modules.

LoRaWAN Gateways: responsible for the reception, demodulation and decoding of the LoRaWAN transmissions. LoRaWAN Gateways relay the aggregated traffic via LTE backhaul.

LoRaWAN Management Network: used for extracting the payload data sent from the gateways. For this project, an open platform, The Things Network (TTN), is utilized to enable public access to community users.

Network Management System (NMS): developed by the industry partner, the platform is called Alpha-X¹. The platform monitors and manages the health of the network and its elements through the control plane. The NMS reduces the management workload and classifies and predicts battery usage based on consumption trends. It also provides tools that allow the user to set up templates of device types and network types to efficiently assign those settings to multiple devices optimizing

the management in large-scale IoT networks. In addition to control plane management, the developed NMS is also responsible for the aggregation of the payload from both the TTN and LTE networks, where the aggregated payload data is forwarded to the data analytic/visualization platform.

Cloud Data Platform: utilized for data storage, analytics, and visualization. The cybersecurity of the LoRa network is managed by TTN, whereas that of the 4G network is managed by the LTE provider. The authentic security of the gateways and devices is managed by Minnovation instead.

DESIGN METHODOLOGY

The main objective of the radio access network design phase is to determine suitable locations for the gateways. Figure 3 illustrates the network design process that was developed in NMSCN. Ideally, each gateway is expected to serve multiple sensors to optimize the cost-effectiveness of the deployment. Hence, optimal gateway locations are those nearest to clusters of sensors. Geographical areas that include clusters of sensors are referred to as sites. To determine the optimal location of the gateway for a given site we propose the following scheme:

Site survey: Performing site visits is a crucial step to identify candidate locations for installing the gateways and sensors. For instance, it is preferred to install the gateway in an open area that maximizes the line-of-sight probability with the sensors [9].

RF Ray Tracing: After identifying the candidate locations, we further optimize the selection based on iterative RF ray-tracing simulations. In addition to the candidate locations, the simulator also incorporates two important geographic data: (i) the terrain data, often called digital elevation model (DEM), and (ii) the building geometry and heights, often referred as the buildings footprint which is a 2.5D data-set.

Coverage Mapping: The coverage of radio service is determined by the ability of the receiver to decode the transmitted message correctly. This is dependent on both the received signal power and the surrounding noise and interference. Thus, in addition to the ray-tracing simulation it is important to get the interference statistics (or at least the average interference power) around the selected sites because the network is operating in the license-free ISM band. See the references cited herewith for further details about interference modeling [10]–[12]. The power level at which an acceptable success rate is maintained is called the sensitivity of the gateway and is dependent on the utilized

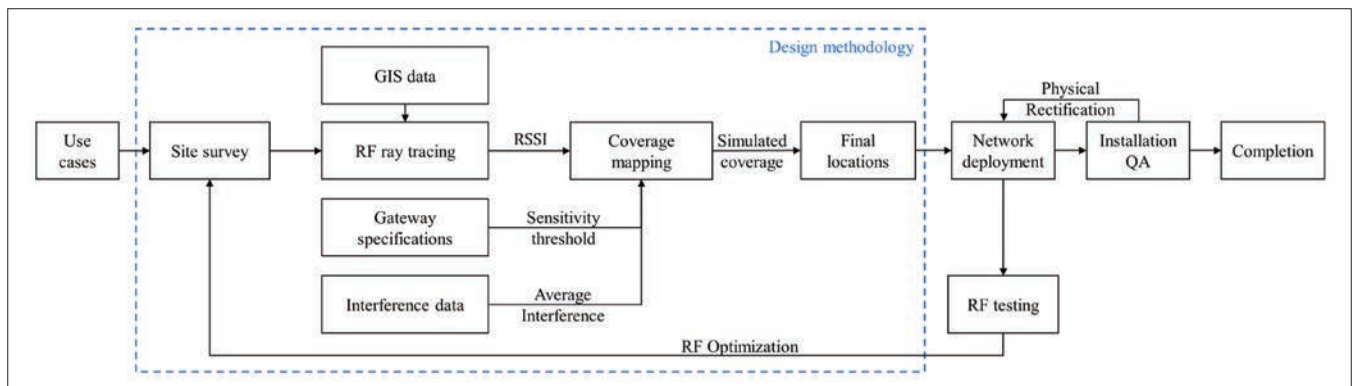


FIGURE 3. Network design methodology utilized in NMSCN to achieve the the design objectives.

spreading factors in LoRa modulation [5]. Higher spreading factors provide better coverage, but require longer time-on-air, which shortens the device's battery lifetime. Hence, it is desirable to utilize the lowest spreading factor to lower the operational costs arising from battery replacement. It is simpler to communicate the outcome of this step as a coverage contour, similar to the one in Fig. 6. Radio coverage design is typically an iterative process since the obtained coverage by simulation is still an estimate and cannot incorporate precisely all real-world factors, including surface materials and foliage, among many more. Thus, some sites may require optimization post-installation using radio frequency (RF) testing, which is further discussed in a later section.

IMPLEMENTATION

The industry partner developed the majority of the IoT devices by integrating commercially available modules and in-house developed firmware. The integration also includes the mechanical housing of IoT devices and the battery management system. The gateways and some sensors required larger rechargeable batteries with solar power system integration. The rooftop installations of the gateways typically require 60 minutes to complete onsite since the gateways themselves are self-contained packages that are specifically designed to minimize site installation. Similarly, each sensor requires around 45 minutes to install since the sensors are also self-contained units, whereas larger sensors that require an external solar panel each needed approximately 1.5 hours to complete.

One of the main challenges of the project is the stringent installation timeline in addition to the COVID-19 pandemic that restricted the typical access to sites and imposed certain work restrictions. Some installations also included hurdles such as sun shading which limited the flexibility for solar panel installations and did not always coincide with the optimal location for coverage. Figure 4 illustrates the implementation phase and highlights examples of the different installation types completed in the network.

The implementation process started with the installation of the gateways first in order for the coverage testing to take place and validate the sensor locations. Accordingly, the results from the testing are utilized to feedback the design phase in order to sustain the gateway connectivity.

COVERAGE VALIDATION AND QUALITY ASSURANCE

RF TESTING AND OPTIMIZATION

The validation of the radio network performance is an important part of the overall verification and quality assurance since only field measurements can reveal the real performance of the network in comparison to the expected performance in the design phase. The proposed methodology utilized for the testing is illustrated in Fig. 5 where we utilize a hand-held

LoRaWAN device referred to as a packet tester and further explained hereafter.

Packet tester: A device we developed at RMIT University in order to facilitate the radio coverage validation. This device sends LoRaWAN frames to the deployed network and logs the properties of these frames including geographic coordinates, spreading factor, transmit power, and frame sequence number. The developed device is shown in Fig. 5 alongside the basic end-to-end architecture of the network testing system. The testing mechanism utilizes similar infrastructure to the gateways deployed along with an additional cloud-based server to extract and store the data received from the testing device via the LoRaWAN network. The packet tester is primarily built using available off-the-shelf modules to achieve better interoperability with common LoRaWAN network server and/or application server. The parameters of testing scenarios are highly configurable, providing the flexibility to configure the LoRaWAN serial keys as well as the spreading factor, transmit power, frequency sub-band, and finally the total number of frames to send for a specific test.

Pin-point test: The first test is to perform pin-point measurements. The packet tester is configured to send a fixed number of frames once it is positioned at the potential locations of the IoT sensors. The ratio of the number of successful frames to the total number of frames is used to estimate the probability of frame success rate. If the probability is higher than a certain threshold, the sensor location is approved for installation. If that is not the case, another location is proposed instead.

Coverage test: The second test is conducted via a drive around the gateway location. The packet tester is configured to continuously transmit LoRaWAN frames while in-motion on a selected route. The drive routes are selected to account for various distances around the gateway as depicted in Fig. 5.

The geographic coordinates are also logged by the packet tester in order to identify the locations from where the frames were transmitted. The transmitter log is extracted from the packet tester, whereas the receiver log is obtained from the cloud server. Both logs are compared in order to identify the lost and successful frames.

To expand the measured coverage to a larger grid, we further use these frames to tune the simulation-based coverage model. This is done by tuning the simulated coverage such that the discrepancy between the simulation and measurements are minimized, in particular by minimizing the root mean square error (RMSE) [13]. The resulting coverage of the gateways in the NMSCN project is depicted in Fig. 6 for data rates DR0 and DR5 (spreading factors SF12 and SF7 respectively). The overall coverage guides the councils to direct future candidate locations of new sensors as well as identifying black spots where the coverage requires enhancement using future gateway installations.

INSTALLATION QUALITY ASSURANCE AND TROUBLESHOOTING

Quality assurance is performed to assess the implementation and ensure the connectivity of the different IoT radio network

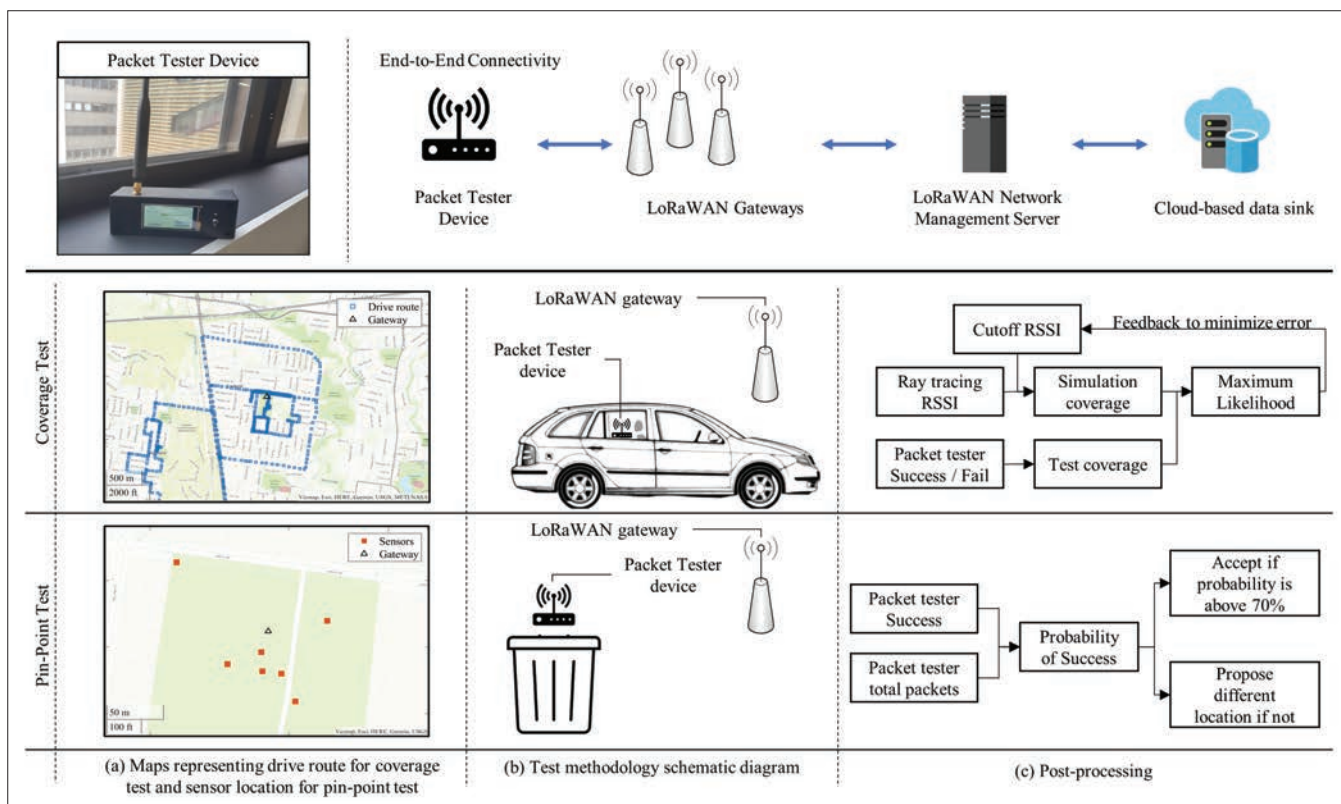


FIGURE 4. Installation photos for all the different network elements included in the NMSCN project.

elements. The quality assurance employed in the NMSCN project is based on two key criteria:

Connectivity status of network elements: In addition to the radio coverage verification, we further test the end-to-end connectivity of gateways and sensors with the NMS. A troubleshooting plan is developed to tackle the connectivity issues that may arise due to coverage limitations or intermittent connection. The plan also includes troubleshooting steps which include tuning the spreading factor, transmit power, frequency sub-band as well as identifying hardware and software deficiencies. If the issue persists, a 4G module is proposed to substitute the LoRaWAN module.

Quality of installation: A procedure conducted to ensure that the physical installation is following the project specifications and the different international and national organizations' requirements. This includes the verticality of the antennas, RF and electrical cabling ratings such as UV protection, as well as battery performance, PV panels' performance, cable conduits, enclosure ratings among many other standards and regulations.

PROJECT MANAGEMENT OF NETWORK DEPLOYMENT

The overall timeline of the project spanned over 18 months to achieve the following key milestones: (i) smart cities network design, (ii) procurement of gateways, sensors and installation services, (iii) implementation of the network elements, (iv) implementation of data visualizations, and (v) broader engagement with the community. The project was managed using PRINCE2 project management methodology and processes. Product-based planning is utilized to identify and document a list of activities to ensure clear communication among the team. Due to the collaboration of eight different project partners, monthly team meetings to communicate the project progress, discuss issues, plan for activities and make decisions were conducted. Furthermore, the goal of this project is to change the councils' conventional methodology from reactive to proactive decision making to provide services which are now driven through data and trends. On the other hand, to raise interest

and awareness, the technical project partners delivered multiple informative workshops followed by targeted meetings to engage the councils. During the final stages of the project, several training sessions have been conducted to provide the council staff with expertise to manage the network prior to handover.

LESSONS LEARNED

This project has provided a great opportunity to get detailed insights both on the technical level and the administrative level when deploying large-scale IoT networks.

TECHNICAL LESSONS

Lesson 1—Conduct a thorough radio planning: When deploying IoT networks in urban environments it is important to validate the marketing claims of many IoT technologies in regards to the distance and performance, as most of the marketing figures do not report the success rate or the level of interference at which the measurements are conducted. Thus, radio planning is a crucial step for successful planning and dimensioning of the networks, especially for large-scale multi-gateway networks. This was one of the key success factors on the project.

Lesson 2—Get the right sensor: With the hype associated with IoT, it is important to frame the targeted outcome from the collected data, because the choice of the type of sensors is crucial in achieving the desired outcomes. For instance, an IoT motion detection sensor still cannot provide occupancy data whereas a sensing camera can. This project leveraged the strength of the industry partner in developing custom-made sensors and was another key success factor.

Lesson 3—Develop a clear quality assurance process: A quality assurance process is a paramount framework aiming to ensure complete implementation, quality installation and reliable connectivity of IoT devices according to the planned use-cases and network design. This process has both technical and financial implications and thus needs to be clearly developed based on industry best practices. In this project, a clear

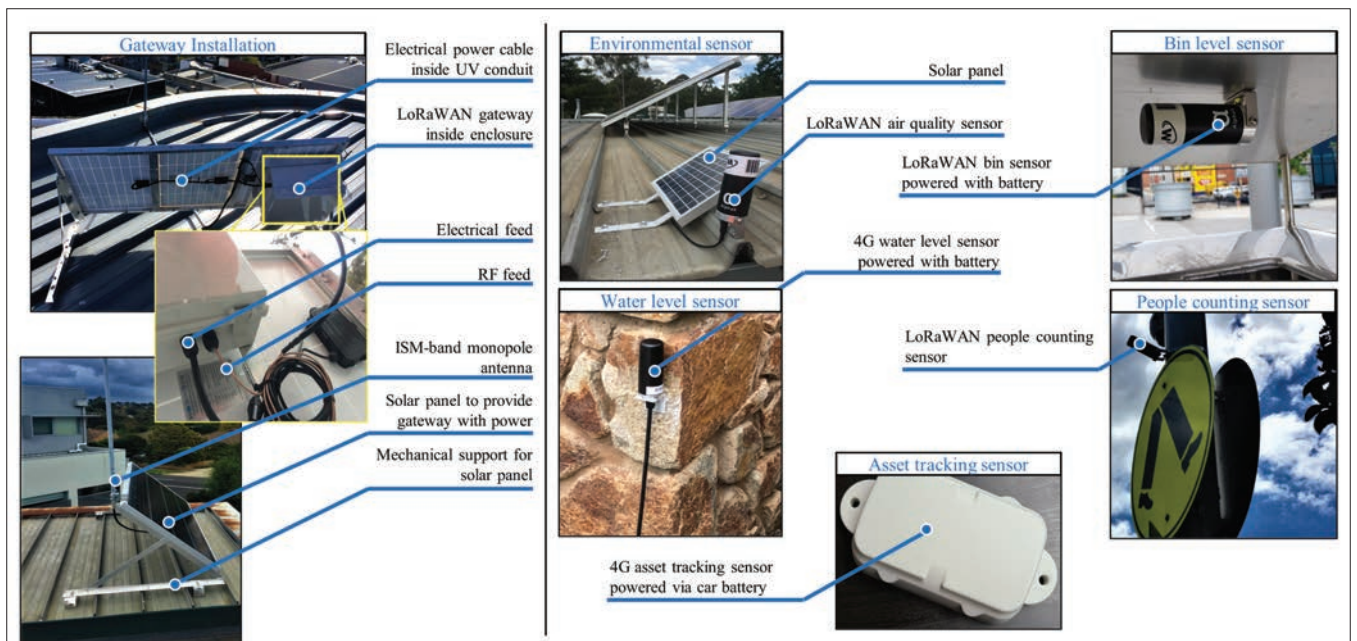


FIGURE 5. An illustration of the packet tester and its architecture and the network performance testing methodology.

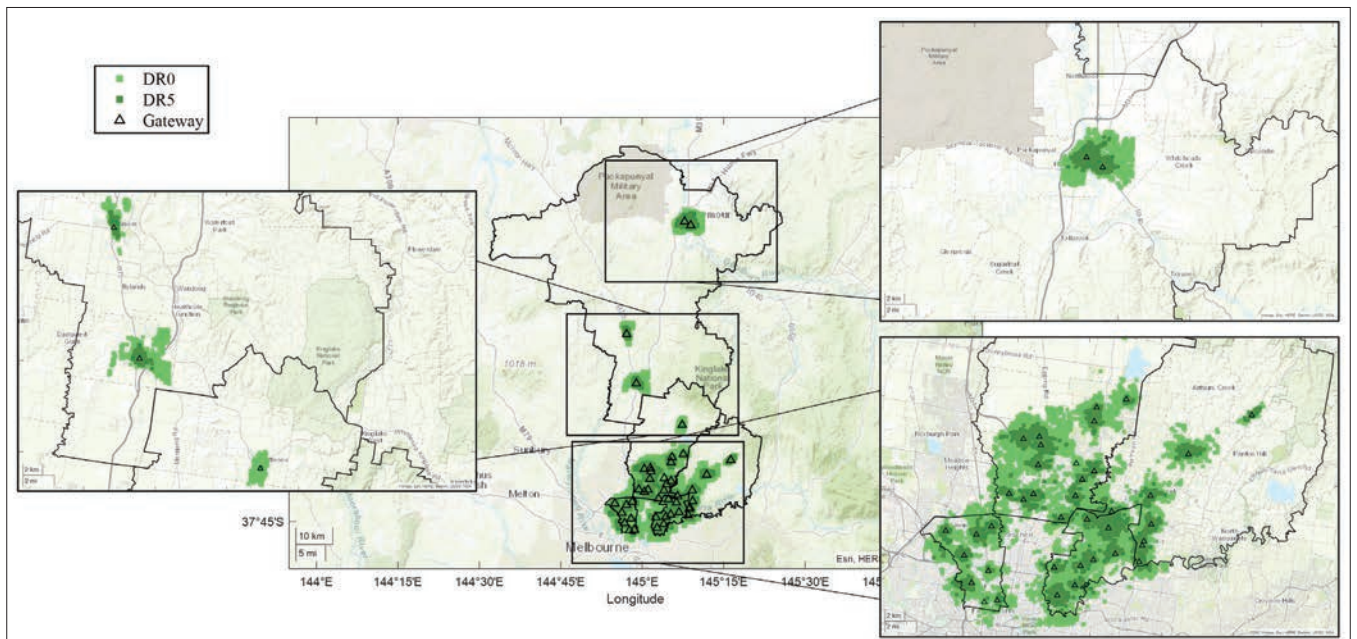


FIGURE 6. Overall NMSCN outdoor coverage map. The light green represents the coverage for DR0 while the dark green represents the coverage for DR5.

QA process was continuously developed and visited that has ensured high quality standards.

ADMINISTRATIVE LESSONS

Lesson 4–End-users’ engagement: The end-user engagement is imperative to enable a more directed and timely project delivery. In this project, the councils were fully engaged, facilitating a seamless transition between different project phases which has led to a timely project delivery.

Lesson 5–Inclusive project management: The importance of having periodical meetings that involve different stakeholders (contractor, network designers, data scientists, and clients) leads to adaptive and robust decision making. This project followed inclusive decision making that has led to its agile roll-out and timely completion.

ADVANCEMENT TO 5G TECHNOLOGIES AND DEPLOYMENT

On one hand 5G, formally known as the IMT-2020 standard, includes three classes of use cases: (i) mobile broadband, (ii) ultra reliable low latency (URLL), and (iii) massive machine-type communications (mMTC) with a focus on telco-deployed networks which sustain a controlled level of interference and quality of service. On the other hand, unlicensed LPWAN such as LoRaWAN particularly complement mMTC use case in 5G by empowering end-users to deploy their own networks (similar to the amalgam of LoRaWAN and 4G deployed in this project). This hybrid approach in having complementary systems, i.e. 5G-mMTC and LPWAN, is significantly accelerating IoT innovation and addresses a wider range of techno-economic needs. This complementary nature is akin to the very successful legacy co-existing of cellular and wireless local area network (WLAN) technolo-

gies that is continuing to play a major roll in last-mile access through 5G and Wi-Fi.

CONCLUSION

The deployment of a large-scale network has almost always been associated with large telecom operators. However, with the introduction of simplified IoT access technologies, the deployment of large-scale networks is now the responsibility of broader entities using free-access radio bands. Therefore, it is crucial to develop a harmonized framework that can facilitate the different aspects of network deployments. In this article, we detailed some of the lessons learned from a large-scale deployment of an open IoT radio access network, where we presented the developed framework for designing, implementing, testing and managing this deployment. We highlighted the different challenges faced in the project on both technical and management levels. Moreover, the NMSCN project attracted two prestigious industry awards from Municipal Association of Victoria (MAV) Technology Awards for Excellence 2020, and IoT Alliance Australia (IoTAA) Smart Cities Award for 2020.

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JAMES DELANEY is currently working toward his master of engineering at the School of Manufacturing, Materials and Mechatronics Engineering, RMIT University, Melbourne, Australia. He received a bachelor of engineering in 2018 also from RMIT University, specializing in electrical and electronic engineering. His current research interests include Internet of Things, smart cities, and machine learning for wireless communications.

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HANS WOLF holds a masters of business (information technology). He is the Chief Information Officer at the City of Whittlesea and sponsor of the Northern Melbourne Smart Cities project. He has led technology teams in various public and private sector organizations ensuring stakeholder value by leveraging technology.

SIMON MASELLI is the founder and CEO of Minnovation Australia (the industry partner in this project). His Australian based company manufactures IoT and telemetry products that have been deployed across Australia and New Zealand, South East Asia and Europe for dozens of global brands.

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KARINA MABELL GOMEZ is a network project manager at Milano Teleport, Italy. In 2015, she served as a senior lecturer at the School of Engineering at RMIT University, where her role was to coordinate several networking courses and supervise several Ph.D. and master students. She obtained her Ph.D. degree in telecommunications from the University of Trento, Italy.

FOOTNOTES

This work was partially funded by the Smart Cities and Suburbs Program, Department of Infrastructure, Transport, Regional Development and Communications, Australian Government. Under the project, Northern Melbourne Smart Cities Network, Enabling Data to Drive Change, grant number SCS69259.

¹ <http://alphax.com.au/>